Deep third-cumulant representations below the dense-Hadamard wall

Mechanistic estimate for the $n = 1024$, $L = 16$ random ReLU MLP; $1\,\mathsf{U} = 2^{31}$ FLOPs, budget $= 1024\,\mathsf{U}$.

Executive answer. The new spectrum measurement kills low-rank approximations of $A \circ P$, but it does *not* kill compression of the *quadratic-form queries* that are actually contracted against the right leg. I would test, in this order: (1) a degree-2 TensorSketch/Hermite feature representation of those queries; (2) an isotropic self-averaging closure for the traceless "noise" part; (3) a diagonal-Hermite version in the shared basis. I do not see a new exact identity that beats V29's $O(n^2 r)$ route without exploiting extra structure. Earlier D_{21}/D_3 slices are not an exact Markov state.

Q1. Representations for deep content

Common rewrite. Let $Q \in \mathbb{R}^{n\times r}$ be the shared old-source basis, q_i^T its ith row, and let $a_j, p_j, f_j \in \mathbb{R}^r$ be column j of the factor matrices for the two left legs and the right leg. A representative term is

$$H = (A \circ P) D_w F^T Q^T = U Q^T, \qquad U_{ic} = \sum_j w_j f_{cj} (q_i^T a_j)(q_i^T p_j).$$

Define

$$S_c = \sum_j w_j f_{cj} a_j p_j^T, \qquad U_{ic} = q_i^T S_c q_i.$$

[DERIVED] Thus the hard object is a family of quadratic forms $\{S_c\}_{c=1}^r$, not the full-rank matrix $A \circ P$ itself. The measured flat spectrum of $A \circ P$ does not determine the effective dimension of this query family.

A. Degree-2 Hermite / TensorSketch queries – my first test

Let $z_i = q_i \otimes q_i \in \mathbb{R}^{r^2}$ and $b_j = a_j \otimes p_j$. With Z containing z_i^T as rows and K containing b_j as columns,

$$U = Z K D_w F^T.$$

Use a TensorSketch/CountSketch $S \in \mathbb{R}^{d\times r^2}$, computable on $a \otimes p$ without materializing the tensor:

$$U \approx \Phi \Psi D_w F^T, \quad \Phi = Z S^T, \quad \Psi = SK.$$

For all old sources, sum the $d \times r$ matrices $\Psi_s D_w F_s^T$ first; multiply by the shared Φ once, sum the four contraction families in $n \times r$, then do one final Q^T multiply. This is an approximate-matrix-multiplication view, not a low-rank approximation of $A \circ P$. TensorSketch is specifically designed to sketch tensor-product features without forming them [1].

[DERIVED] Ignoring the already-existing compressed-factor transport, the contraction cost for m old sources and $t = 4$ families is

$$C_{\text{sk}} \simeq t(m+1)\frac{dr}{n^2} + \frac{r}{n} \quad \mathsf{U}/\text{layer},$$

with sketch construction below about $0.05\,\mathsf{U}$ here. At layer 14 ($m = 10$):

	$d = 64$	$d = 128$	$d = 256$
$r = 224$	0.82	1.42	2.63
$r = 384$	1.41	2.44	4.50

If factor transport is two $r \times n$ full legs per source, add roughly $2mr^2/n^2\,\mathsf{U}$ (uniform-r upper estimate: $0.96\,\mathsf{U}$ at $r = 224$, $2.81\,\mathsf{U}$ at $r = 384$). The actual nested-rank mix is smaller.

[CONJECTURE] What is lost is only the quadratic-feature component orthogonal to the sketch. Flat singular values make $d = 64$ aggressive, but aggregation over sources and the low-rank right factor may reduce the required d drastically.

[MEASURE] First falsifier: on the saved layer-14 legs, compute exact *aggregate old* U before the final Q^T. Try $d = 32, 64, 128, 256, 512$, 10 seeds. If median relative RMS error is $> 2\%$ at $d = 128$ (or has a heavy $> 3\%$ tail), this will not reach leader cost. This is one-network BLAS plus sketches; no forward Monte Carlo is needed.

B. Self-averaged trace closure for the noise-like part

For an old, mixed basis, model a row as isotropic conditional on its leverage $\ell_i = \|q_i\|^2$:

$$\mathbb{E}[(q_i^T a)(q_i^T p) \mid \ell_i] \approx \frac{\ell_i}{r} a^T p.$$

Then

$$\bar{U}_{ic} = \frac{\ell_i}{r} \sum_j w_j f_{cj} (a_j^T p_j).$$

This carries only the factor columns plus the leverage vector; no dense Hadamard product. **[DERIVED]** Runtime contraction is $O(nr)$ per source/family plus one shared $n \times r$ by $r \times n$ multiply, so the new contraction cost is essentially $r/n = 0.219\,\mathsf{U}$ ($r = 224$) or $0.375\,\mathsf{U}$ ($r = 384$) per layer, plus compressed-factor transport.

A useful variance diagnostic follows from Wick's rule. For Gaussian-isotropic q,

$$\mathrm{Cov}(z_j, z_k) \propto (a_j^T a_k)(p_j^T p_k) + (a_j^T p_k)(p_j^T a_k), \quad z_j = (q^T a_j)(q^T p_j).$$

[CONJECTURE] If the traceless quadratic part is truly noise, its contraction with F^T may concentrate even though $A \circ P$ is full-rank. If instead the target is dominated by quenched anisotropy, this closure dies immediately.

[MEASURE] First falsifier, including the requested resampling test: choose one age-7–10 source. Freeze its birth state and resample the later weight suffix 8–16 times; compute its exact U at one target layer. Let μ be the resampled mean and σ_{cond} the RMS deviation. The fixed-network target can be replaced by expectation only if $\sigma_{\mathrm{cond}}/\|\mu\|_F \lesssim 0.02$ (also test the *aggregate* old sum, where cancellation can help). A ratio > 0.1 falsifies self-averaging decisively.

C. Diagonal Hermite closure in Q coordinates

Keep the coordinate-square Hermite features but drop cross-coordinate products. Approximate each S_c by its diagonal in the current shared basis:

$$U \approx (Q \circ Q)D, \qquad D = (F_A \circ F_P) D_w F^T \in \mathbb{R}^{r \times r}.$$

[DERIVED] This never forms A, P, or $A \circ P$. For m sources and four families, after aggregating the D_s matrices,

$$C_{\mathrm{diag}} \simeq 4(m+1)\frac{r^2}{n^2} + \frac{r}{n} \quad \mathsf{U}/\text{layer}.$$

At $m = 10$: $2.32\,\mathsf{U}$ for $r = 224$, $6.56\,\mathsf{U}$ for $r = 384$, again plus factor transport. It loses all off-diagonal quadratic Hermite modes. A common rotation that approximately jointly diagonalizes sampled S_c could improve accuracy, but rotating every source factor costs about $r^2/n^2\,\mathsf{U}$ per full leg and must earn that cost.

[MEASURE] First falsifier: from stored factors, compute exact aggregate U and this diagonal approximation at layers 10 and 14. If error is $> 5\%$, try one common rotation fitted from 4–8 random linear combinations $S(y) = \sum_c y_c S_c$. If the best rotated diagonal remains $> 2\%$, discard this route.

D. Can earlier D_{21}/D_3 slices be an exact state?

Let K be a symmetric third-cumulant tensor and L the next dense linearized map. With $v_a = L^T e_a$,

$$D'_{21}(a, c) = K(v_a, v_a, v_c).$$

[DERIVED] The current D_{21} gives only $K(e_i, e_i, e_c)$, so it is insufficient for generic dense v_a. An exact *one-step* sufficient state is

$$S_L(a, k) = K(v_a, v_a, e_k) \in \mathbb{R}^{n \times n}, \qquad D'_{21} = S_L L^T.$$

But closure for arbitrary future maps requires the quadratic map $q(v) = K(v, v, \cdot)$, with $n\,n(n+1)/2 = \Theta(n^3)$ parameters. Therefore there is no exact slice-only Markov state of $O(n^2)$ size without extra structure. Precomputing only p future slices and then dropping the source is just a delayed window and is covered by your dead window result.

A genuinely new version would need a *history-dependent aggregate response closure* that regenerates future-pulled slices from extra matrix state. One or two $n \times n$ response matrices would cost roughly 1–2 U/layer. **[MEASURE]** Falsify it by measuring whether $K(v, v, \cdot)$ for 64 random dense v can be predicted to 2% from (C, D_{21}, D_3) plus one previous response matrix. Your $+0.02R^2$ result for scalar/live features does not test this high-dimensional equivariant state.

E. Exact identity question

[DERIVED] The useful exact association is already

$$H = \left[((QF_A) \circ (QF_P))D_w F_A^T\right]Q^T,$$

which is $O(n^2 r)$ once the dense left evaluations exist and is essentially V29's re-contraction. Avoiding those evaluations turns the problem into multiplying the row-Khatri–Rao matrix $[q_i \otimes q_i]^T$ $(n \times r^2)$ by the column-Khatri–Rao matrix $[a_j \otimes p_j]$ $(r^2 \times n)$. With $r^2 \geq n$ and your observed full numerical rank, low rank alone supplies no exact shortcut. I see no generic exact $O(nr^2)$ or lower identity; any such win must exploit additional structure such as diagonalizability, concentration, or sketchability tested above.

Q2. Three hypotheses for the leaders, ranked

1. Aggregate response state (highest prior). **[CONJECTURE]** They may sum old-source history into 1–2 dense response matrices updated recursively, rather than retain source identities. This uniquely matches "young-tier cost plus almost nothing" and can preserve quenched information that a memoryless slice closure loses. **Distinguishing signature:** cost becomes $O(L)$ after the age cutoff, not $O(L^2)$ in source-layer pairs; deterministic error; increasing depth from 16 to 24 should add roughly a constant 1–2 units/layer, not an expanding old tier.

2. Quadratic-feature sketch of old contractions. **[CONJECTURE]** This is Proposal A with $d \approx 64$–128 and aggressive aggregation across sources/families. **Signature:** a smooth accuracy-versus-d curve and seed-to-seed variation; local D21 error is nearly zero-mean. At $d = 64$, layer-14 contraction cost is only $0.82\,\mathsf{U}$ for rank 224 before factor transport.

3. Trace/diagonal Hermite closure of the old residual. **[CONJECTURE]** Leaders may keep only the coherent low-order quadratic modes and treat the flat remainder statistically. The trace version costs about $0.22\,\mathsf{U}$/layer at $r = 224$ (plus factor transport); the diagonal version is about $2.32\,\mathsf{U}$/layer at $m = 10, r = 224$. **Signature:** essentially flat or weakly growing cost versus number of old sources, deterministic results, and a sharp correlation between success and the fraction of U explained by the isotropic/diagonal quadratic component – not by the singular spectrum of $A \circ P$.

Q3. If these fail: the next single measurement

[MEASURE] Measure the **quadratic-query compression curve** on one layer-14 network: for the *aggregate old contribution before* Q^T, record relative RMS error for (i) isotropic trace only, (ii) diagonal Hermite, and (iii) TensorSketch widths $d = 16, 32, 64, 128, 256, 512$. This is the right spectrum to measure now. It directly returns the minimum quadratic-feature dimension $d_{2\%}$ actually seen by D_{21}. If $d_{2\%} \leq 128$, a leader-cost old tier is credible; if it is > 512, both self-averaging and sketch-based explanations are largely dead, and I would concentrate on an aggregate response recursion rather than another tensor factorization.

Reference. [1] N. Pham and R. Pagh, "Fast and Scalable Polynomial Kernels via Explicit Feature Maps," KDD 2013, DOI: 10.1145/2487575.2487591. TensorSketch computes sketches of tensor products without explicitly materializing the tensor.